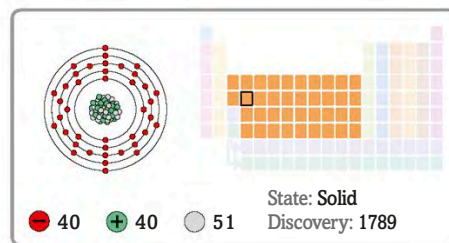


40
Zr

Zirconium



Forms



This greyish-white pure element is easy to shape.

Bar of pure zirconium refined in a laboratory



Uses

Dental crowns



These tough dental crowns are made of a zirconium-rich ceramic.

Ceramic knife



This hard, non-metallic blade does not require frequent sharpening.



Camera flash from the 1960s

This zirconium-filled bulb produces a bright light.

Zirconia crystal ring



This ring contains cubic zirconia crystals.

This element is named after the mineral **zircon**, which means “golden” in Persian, a reference to the golden-brown colour of its **crystals**. The Swedish chemist Jacob Berzelius was the first person to isolate **pure zirconium**, in 1824. Today, however, the element is mostly used

in the form of the compound zirconium dioxide, or **zirconia**. Powdered zirconia is heated to produce a hard glass-like ceramic, which is used to create **dental crowns** and sharp **ceramic knives**. Powdered zirconia also forms sparkling **zirconia crystals** that look like diamonds.